

Problem A. Cartas

Time Limit 2000 ms

Problem Statement

There are N cards, numbered 1 to N .

Card i has a positive integer A_i written on it.

You can choose three of these cards and concatenate the integers written on them in any order you like to make a new integer. For example, if you choose cards with 1, 23, and 4 written on them, you can make integers such as 1234 and 4231.

Find the maximum integer you can make.

Constraints

- $3 \leq N \leq 2 \times 10^5$
- $1 \leq A_i < 10^6$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
 $N$   
 $A_1 A_2 \dots A_N$ 
```

Output

Print the answer.

Sample 1

Input	Output
5 1 4 3 5 8	854

If you choose cards with 4, 5, and 8 written on them, you can make 458, 485, 548, 584, 845, or 854.

You can make nothing greater than 854, so the answer is 854.

Sample 2

Input	Output
8 813 921 481 282 120 900 555 409	921900813